Seminar Paper

Urban Land-Use Efficiency

A Stable Metric for SDG 11.3.1 and Its Economic Drivers

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Abstract

The official SDG 11.3.1 indicator, the ratio of land consumption rate to population growth rate (LCRPGR), is unstable: it divides by a growth rate that approaches zero and mixes an arithmetic numerator with a logarithmic denominator. We propose the Built-up per Capita Rate (BpCR), the log change in built-up area per person, which equals the difference of two log growth rates and is therefore well-defined for every country-period. Using a harmonised global panel of 193 UN member states (1985–2020) built from GHSL built-up and population grids under the Degree of Urbanisation, we (i) document where LCRPGR breaks down, (ii) show BpCR reproduces the policy signal without the instability, and (iii) estimate its economic drivers with two-way fixed-effects and dynamic-panel GMM models.

Keywords: SDG 11.3.1, Urban land-use efficiency, Built-up per capita rate (BpCR), LCRPGR, Dynamic panel GMM, GHSL / Degree of Urbanisation

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1. Introduction

The world is urbanising rapidly: by the middle of this century close to two-thirds of humanity will live in cities, and almost all of the projected growth in built-up land will occur in developing countries (Gao and O'Neill, 2020). The scale of expansion is striking: every year the world's built-up area grows by roughly the size of the Netherlands (Seto et al., 2012). For development policy the central question is not whether cities expand, but how efficiently they do so, that is, how much additional land each new urban resident consumes. When built-up area grows in step with population, cities densify and remain compact; when it grows much faster than population, they sprawl. Sprawl carries real costs. Once farmland on the urban fringe is built over, it is effectively lost for good. Low-density, spread-out development makes people more dependent on cars, because homes, jobs, and services lie far apart. It also makes public infrastructure more expensive per resident, since the same roads, pipes, and power and transport networks must reach a smaller number of people over a larger area. And by encouraging longer car journeys and land-hungry construction, sprawl tends to raise greenhouse-gas emissions per person. Measuring land-use efficiency well is therefore a precondition for monitoring sustainable urbanisation.

The United Nations monitors this through target 11.3 of its Sustainable Development Goals (SDGs), whose official indicator 11.3.1 is the ratio of the Land Consumption Rate to the Population Growth Rate (LCRPGR) (UN-Habitat, 2025). Because LCRPGR is the agreed global yardstick, its statistical properties are not a technical detail: they determine how every country's progress is judged and compared. Yet the indicator is fragile by construction. It divides an arithmetic land-consumption rate by a logarithmic population-growth rate, so it is internally inconsistent; and because the population-growth rate sits in the denominator, the ratio diverges towards plus or minus infinity and changes sign whenever population is roughly stable. In exactly the countries where land-use efficiency matters most for policy, the official metric becomes uninformative.

This paper makes three contributions. First, it documents, on a harmonised global dataset, that LCRPGR is statistically unstable and that its instability is concentrated where population growth is near zero. Second, it proposes a simple, log-based alternative, the Built-up per Capita Rate (BpCR), defined as the logarithmic growth rate of built-up area per capita. BpCR is the difference of two logarithmic growth rates, which makes it finite for every country-period, symmetric between densification and sprawl, additively decomposable, and directly interpretable; it reproduces the policy signal of LCRPGR where the latter is well-behaved, and so complements rather than replaces the official indicator. Third, using the stable metric, the paper estimates the economic and structural drivers of per-capita urban land use on a panel of 193 United Nations member states observed at five-year intervals between 1985 and 2020.

The empirical analysis builds the panel from satellite-based settlement data under the global Degree of Urbanisation standard, so that built-up area and population are measured consistently. To address the well-known downward bias of the lagged dependent variable in dynamic fixed-effects models, the persistence of per-capita land use is estimated with Arellano-Bond and Blundell-Bond generalised method of moments (GMM) estimators. Three results stand out. Sprawl is the global norm rather than the exception: built-up area per capita rises in a clear majority of country-periods. Per-capita land use is strongly path-dependent, with about three-fifths of one period's value carried into the next. And within countries, denser cities, net in-migration, and higher income are all associated with less land consumed per resident, indicating that, over time, economic growth is accompanied by densification rather than sprawl.

The remainder of the paper is organised as follows. Section 2 sets out the SDG 11.3.1 indicator, its structural limitations, and the BpCR alternative. Section 3 reviews the related literature. Section 4 describes the data and Section 5 presents descriptive evidence. Section 6 details the empirical strategy, Section 7 reports the results, and Section 8 gives robustness checks. Section 9 discusses the findings and Section 10 concludes.

2. Measurement Framework

This section sets out the conceptual core of the paper. We first define the official SDG 11.3.1 indicator and show why it is statistically unstable, and we then introduce the stable, log-based alternative used in the empirical analysis.

2.1. The SDG 11.3.1 Indicator and Its Limitations

Following the UN-Habitat metadata (UN-Habitat, 2025), the official indicator divides an arithmetic land consumption rate by a logarithmic population growth rate:

$$\mathrm{LCRPGR} = \frac{\mathrm{LCR}}{\mathrm{PGR}}, \qquad \mathrm{LCR} = \frac{V_t - V_{t-z}}{V_{t-z}} \cdot \frac{1}{z}, \qquad \mathrm{PGR} = \frac{\ln(P_t/P_{t-z})}{z},$$

where the symbols are defined as follows: V_t is the urban built-up area (in square kilometres) at the end of the period and V_{t-z} the built-up area at the start; P_t and P_{t-z} are the corresponding urban populations; z is the length of the period in years; $\ln(\cdot)$ denotes the natural logarithm; and the subscript i indexes the country. The Land Consumption Rate (LCR) is thus the arithmetic growth of built-up area, expressed per year, and the Population Growth Rate (PGR) is the logarithmic (continuous) growth of population per year; the indicator LCRPGR is their ratio. Two structural problems follow. **(1) Division by a near-zero denominator.** When population is roughly stable (PGR ≈ 0) the ratio diverges towards $\pm\infty$ and flips sign, so small, ordinary demographic differences produce wild, uninterpretable values. **(2) Arithmetic-versus-logarithmic asymmetry.** The numerator is an arithmetic growth rate while the denominator is logarithmic, so the ratio is not symmetric and does not aggregate cleanly over time (Nicolau et al., 2019). The result is a metric whose tails are dominated by an arithmetic artefact rather than by real land-use behaviour.

2.2. A Stable Metric: The Built-up per Capita Rate (BpCR)

The UN-Habitat metadata itself concedes that the ratio is hard to interpret and offers built-up area per capita (V/P) as a secondary indicator (UN-Habitat, 2025). That quantity is a level, a snapshot at one point in time. Following Lu and Weng (2026), we turn it into a trend by taking its annual growth in logarithmic form:

$$\mathrm{BpCR}_{i,t} = \frac{1}{z} \ln\left(\frac{V_t/P_t}{V_{t-z}/P_{t-z}}\right) = \mathrm{LCR}_{\log} - \mathrm{PGR}_{\log}.$$

Because BpCR is the **difference of two logarithmic growth rates**, it is (i) finite and well-defined for every country-period (no division by ~0), (ii) symmetric (densification and sprawl are mirror images around zero), (iii) additively decomposable, and (iv) directly interpretable: BpCR > 0 means sprawl (more land per resident), BpCR < 0 densification, and BpCR $= 0$ land growing exactly in step with population. BpCR reproduces the policy signal of LCRPGR where the latter is well-behaved, but without the instability; it complements rather than contradicts the official indicator.

3. Literature Review

The literature on urban land use splits into two strands. One asks what *drives* urban expansion; the other scrutinises how land-use efficiency is *measured*. This paper sits at their intersection.

3.1. Drivers of Urban Land Expansion

Seto et al. (2011) synthesise 326 remote-sensing studies of global urban land expansion between 1970 and 2000. They document that urban land has grown faster than urban population almost everywhere, and that the expansion is driven mainly by income in most regions and by population growth in India and Africa. Oueslati et al. (2015) move from description to econometrics, estimating the determinants of sprawl on a panel of 282 European cities (1990, 2000, 2006) within a monocentric-city framework. They find that rising income and population both enlarge the urban footprint, and that cities keep spreading even where population is flat. Together these studies establish income and population as the leading drivers of urban land use, but they work mostly across cities or countries and at a single cross-section, leaving open whether the same relationships hold within a country as it develops over time.

3.2. Critiques of the Indicator: Formula and Data

Nicolau et al. (2019) assess the SDG 11.3.1 formula itself, comparing the official LCRPGR with a land-use-efficiency-derived alternative at the municipality level in mainland Portugal. They conclude that the LCRPGR formula is structurally unstable and that a logarithmic, efficiency-based alternative is more robust. Lu and Weng (2026) broaden the question to data sensitivity, evaluating land-use efficiency across six built-up and four population datasets for 176 countries over 2000–2020. They show that efficiency estimates swing widely with the choice of dataset and argue that a log-based framework is needed for reliable monitoring. This strand establishes that the choice of metric, and not only the choice of data, can change the conclusions drawn about sustainable urbanisation.

3.3. Research Gap and Contribution

Our contribution sits at the intersection of the two strands. On measurement, we adopt the log-based metric that both Nicolau et al. (2019) and Lu and Weng (2026) recommend, in the form of BpCR. On method, we make it dynamic: a two-way fixed-effects panel with Arellano-Bond and Blundell-Bond GMM, estimated on 193 United Nations member states over 1985–2020, that measures the persistence (path-dependence) of per-capita land use. The result refines the drivers literature: while richer countries look more sprawled in the cross-section, within a country growing richer is associated with densification rather than sprawl.

4. Data

4.1. Data Sources

The empirical analysis draws on three official, global data providers, the same sources used in international SDG monitoring. Built-up area, population, and the urban mask come from the **Global Human Settlement Layer (GHSL)** of the European Commission's Joint Research Centre (Pesaresi et al., 2024; Schiavina et al., 2023). Within GHSL we use three products: GHS-BUILT-S for built-up surface, classified from Landsat and Sentinel-2 imagery by machine learning; GHS-POP, which disaggregates census counts onto that built-up surface; and GHS-SMOD, which applies the Degree of Urbanisation (DEGURBA) method to flag urban cells. The layers are available at five-year epochs; we use the observed epochs from 1985 to 2020 and discard the 2025 and 2030 modelled projections. Administrative units come from the **Global Administrative Unit Layers (GAUL)** of the UN Food and Agriculture Organization (FAO, 2024), which provide UN-validated, politically neutral country boundaries with global coverage. Socio-economic controls come from the **United Nations**: GDP per capita in constant 2020 US dollars from UN SNAAMA and net migration as a share of population from UN DESA World Population Prospects 2024 (UN SNAAMA, 2025; UN DESA WPP, 2024).

Table 1 summarises the technical characteristics of each dataset.

Table 1: Dataset specifications. GHSL grids are distributed in the World Mollweide equal-area projection (ESRI:54009); built-up and population are aggregated at 100 m and the SMOD urban mask at 1 km. Vector and tabular sources are listed with their native form.

Dataset	Provider	Type	Resolution	CRS	Coverage	Variable
GHS-BUILT-S R2023A	EC JRC	Raster	100 m	Mollweide (54009)	1975-2030, 5-yr	Built-up area (m2/cell)
GHS-POP R2023A	EC JRC	Raster	100 m	Mollweide (54009)	1975-2030, 5-yr	Population (persons/cell)
GHS-SMOD R2023A	EC JRC	Raster	1 km	Mollweide (54009)	1975-2030, 5-yr	Degree of Urbanisation (class 10-30)
GAUL 2024/2025	FAO	Vector	Levels 0-2	WGS84 (EPSG:4326)	Global	Admin boundaries
UN SNAAMA	UN Stats	Tabular	Country-year	–	1970-2024	GDP p.c. (const. 2020 US$)
UN DESA WPP 2024	UN DESA	Tabular	Country-year	–	1950-2023	Net migration (CNMR /1000)

4.2. Constructing the Panel Following the UN Metadata

A central design choice is that we build the panel exactly as the SDG 11.3.1 metadata prescribes (UN-Habitat, 2025), so that our BpCR and the official LCRPGR are computed on identical inputs and the comparison is like-for-like. The construction follows the metadata stages.

Figure 1 illustrates the construction for one example window, with each layer drawn over a satellite basemap. Panel (a) is the basemap for reference; panel (b) shows the GHS-BUILT-S surface as the built-up share of each cell; panel (c) the GHS-SMOD Degree of Urbanisation, which classifies each cell from rural to urban centre; and panel (d) the urban-masked built-up that actually enters the analysis, retaining built-up only in cells classified as urban (SMOD at least 21). The same masking is applied to population, so that both components are measured on a consistent urban footprint.

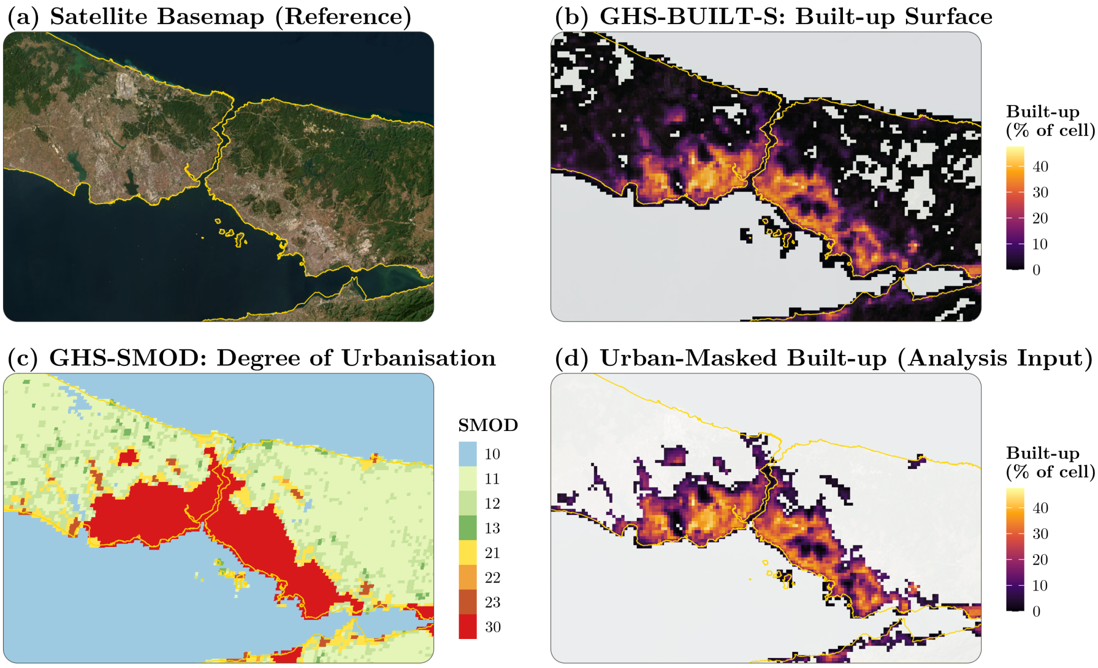

Figure 1: Building the analysis input following the SDG 11.3.1 metadata, for the Istanbul / Bosphorus window, with each layer overlaid semi-transparently on a satellite basemap. (a) Satellite basemap (Esri) for reference. (b) GHS-BUILT-S built-up surface, shaded by the share of each 1 km cell that is built up (0 to about 50%). (c) GHS-SMOD Degree of Urbanisation, which classifies every cell from water and rural (blues and greens) through suburban and town classes to the urban centre (red); cells coded 21 or higher count as urban. (d) Urban-masked built-up, the input actually used: the built-up surface of panel (b) kept only where panel (c) is urban, on the same built-up colour scale. The gold line is the GAUL country boundary (here the coastline).

Analysis area. SDG 11.3.1 is, by construction, an urban (city-scale) indicator: it is measured inside cities rather than over whole national territories. We delineate the urban area with DEGURBA on the GHS-SMOD grid, classifying a cell as urban when its settlement code is at least 21 (urban centres and dense or semi-dense towns and their suburbs). All built-up and population quantities are measured only over these urban cells.

Components and aggregation. Our unit of analysis is the country, so we aggregate the urban cells to the national level. Crucially, we aggregate the *components*, not the metric: for each country and five-year period we sum urban built-up area V and urban population P over all urban cells to obtain national urban totals, and only then form the rates. This follows the metadata's aggregate-then-rate rule, which warns against averaging ratios across sub-units. Averaging city-level ratios would let a small town with a near-zero population denominator dominate the national figure; summing the components first avoids this and keeps the national rate well-defined.

Rates and indicators. From these national totals we compute the Land Consumption Rate (arithmetic growth of built-up area per year) and the Population Growth Rate (logarithmic growth of population per year), their ratio LCRPGR, and the built-up-per-capita level. From the same totals we also compute BpCR as the logarithmic growth of built-up area per capita, as defined in Section 2.

Control variables. The two urban controls are our own calculations from the same GHSL/SMOD layers, not external series. Urban land area is the area of the cells classified as urban (GHS-SMOD code at least 21), aggregated to the national level from the SMOD grid; **urban density** is then national urban population divided by this urban land area, and we use it in logs. **Urban population share** is national urban population (over urban cells) divided by national total population (over the whole territory). The remaining two controls come directly from international sources rather than being constructed here: **GDP per capita** (constant 2020 US dollars, in logs) from UN SNAAMA, and **net migration** as a share of population from UN DESA World Population Prospects 2024. All variables are aligned to the same country-period grid as the built-up and population components.

The resulting dataset is a balanced-where-observed panel of 193 United Nations member states across eight five-year epochs from 1985 to 2020, giving 1,538 country-periods with an observed BpCR. The dynamic regressions in Section 7 use the subset with the lagged dependent variable and all four controls present (complete-case estimation), which reduces the sample to roughly 1,500 country-period observations.

5. Descriptive Statistics

Even before any regression, the raw data already reveal why the choice of metric matters. On the estimation window (1538 country-periods, 1985–2020), the official log-LCRPGR has a standard deviation of about 18 and ranges from roughly -568 to 281: its division by a near-zero population-growth rate produces enormous, sign-flipping outliers. BpCR, by contrast, stays in a tight band with a standard deviation of about 0.017, roughly 1,020 times narrower. The two metrics describe the same underlying built-up and population data, yet one is dominated by an arithmetic artefact and the other is not.

Sprawl is the global norm rather than the exception. BpCR is positive in 63% of country-periods, and 129 of 193 countries (67%) have a positive average BpCR over the period, meaning built-up area per capita is rising: cities are consuming more land per resident. The median country-period BpCR is +0.0041, and the official LCRPGR exceeds one (its sprawl threshold) in 56% of cases. The qualitative message is the same under either metric; the difference is that BpCR delivers it without the numerical instability.

Table 2: Summary statistics, 1985-2020 (country-periods with an observed BpCR). LCRPGR is the official log ratio; BpCR is the log growth of built-up area per capita.

Variable	N	Mean	SD	Min	Median	Max
BpCR	1538	0.004	0.017	-0.114	0.004	0.110
LCRPGR (log)	1538	1.304	17.532	-568	1.031	281
ln(Urban density)	1538	7.581	0.572	5.719	7.606	9.618
Urban pop. share	1538	0.673	0.167	0.122	0.702	1.000
ln(GDP per capita)	1501	8.394	1.517	4.795	8.298	12.139
Net migration (% pop)	1538	-0.112	2.268	-70.990	-0.043	8.622

Table 2 reports summary statistics for the dependent variables and controls over the estimation window.

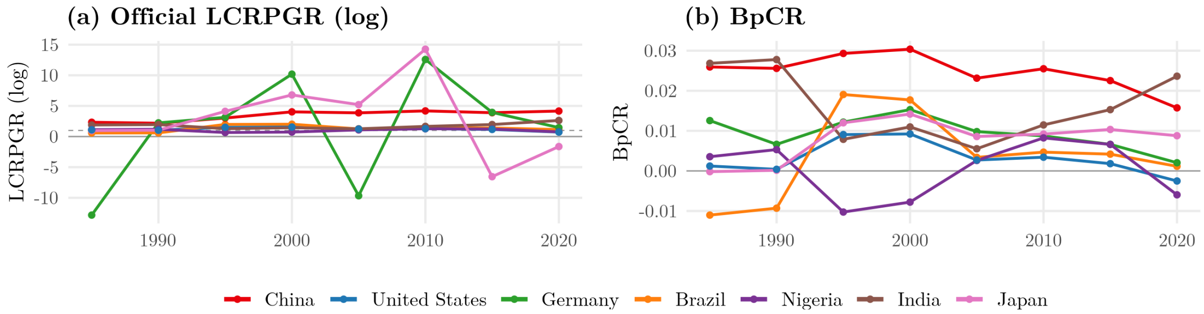

Figure 2: Country trajectories for seven regional representatives, 1985–2020. (a) The official log-LCRPGR swings erratically and crosses its zero and unity reference lines unpredictably; (b) BpCR is smooth and interpretable on the same data, with the fastest-urbanising economies showing the strongest early sprawl before easing.

Country trajectories make the contrast concrete (Figure 2). For a set of regional representatives, the official LCRPGR swings erratically from one period to the next, crossing zero and spiking to large positive and negative values that are hard to read as land-use behaviour. The BpCR series for the same countries are smooth and interpretable: most drift gradually towards balance, while the fastest-urbanising economies show the strongest sprawl earlier in the period before easing. The same data, read through a stable metric, become legible.

6. Empirical Strategy

Research question. We ask two linked questions. First, a substantive one: what economic and structural factors drive *within-country* changes in per-capita urban land use over time, and how persistent are those changes? Second, a measurement one: does the official SDG 11.3.1 indicator capture this reliably, or does the choice of metric change the conclusions? We formalise these in five hypotheses (Table 3).

Table 3: Hypotheses

	Hypothesis	Expected
H1	BpCR captures land-use efficiency more reliably than LCRPGR	better fit
H2	Higher urban density is associated with less land per capita	−
H3	Higher income is associated with less land per capita	−
H4	Net in-migration densifies cities	−
H5	Per-capita land use is path-dependent	+

6.1. Specification

We estimate a single specification, run once with each metric as the dependent variable so that the comparison is strictly like-for-like:

$$\text{Metric}_{i,t} = \rho\,\text{Metric}_{i,t-1} + \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2\,\text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4\,\text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t},$$

where Metric is either LCRPGR or BpCR for country i in period t. The country fixed effects μ_i absorb all time-invariant national characteristics (geography, institutions, the initial level of urbanisation), so identification comes from changes *within* a country over time rather than from comparisons *between* countries. The period fixed effects δ_t absorb global shocks common to all countries in a given epoch (technology, the global business cycle, and any epoch-specific features of the satellite record). Standard errors are clustered by country to allow for serial correlation within a country's series. The four regressors are chosen because each has a clear theoretical link to how much land a city consumes per resident. **Urban density** captures the compact-city mechanism: where settlement is already dense there is both less room and less need to expand outward, so additional residents are accommodated within the existing footprint, implying less new land per capita (H2). **GDP per capita** has an ambiguous sign a priori. Across countries, higher income has historically gone with more sprawl, through larger homes and car-based mobility; but within a country, rising income can also fund vertical and infill construction, redevelopment, and stronger land-use regulation, which densify rather than expand the footprint. Our within-country design tests which force dominates over time (H3). **Net migration** proxies population pressure: in-migrants typically settle in the existing urban stock, so housing demand is absorbed by densification faster than the built-up area grows (H4). Finally, the **lagged dependent variable** captures path-dependence: the building stock, road network, and zoning that shape land use are long-lived, so a country's per-capita land use tends to persist from one period to the next (H5). H1 concerns the metric itself and is addressed by comparing the LCRPGR and BpCR runs.

6.2. Two Stages: Static and Dynamic

We proceed in two stages. In the **static** stage we drop the lagged dependent variable and estimate the two-way fixed-effects model with each metric. This isolates the role of the *metric*: with the data, controls, and fixed effects held identical, any difference between the LCRPGR and BpCR results is attributable to the dependent variable alone (H1). In the **dynamic** stage we add the lagged dependent variable to measure path-dependence (H5), that is, the extent to which a country's per-capita land use in one period carries into the next.

6.3. Dynamic-Panel Estimation and the Nickell Bias

Adding a lagged dependent variable to a fixed-effects model introduces a well-known problem. The within (demeaning) transformation that removes μ_i makes the transformed lagged regressor mechanically correlated with the transformed error, biasing the estimate of ρ downward; the bias is of order $1/T$ and is therefore substantial in a panel with only eight periods such as ours. To correct it we use generalised method of moments (GMM) estimators designed for dynamic panels.

The **Arellano-Bond** (difference) estimator first-differences the equation to eliminate the country fixed effects and instruments the differenced lagged term with deeper lags in levels. The **Blundell-Bond** (system) estimator augments this with a second, levels equation in which the lagged term is instrumented by its own past differences; this added moment information makes it more efficient when the series is persistent, as BpCR is. We treat **Arellano-Bond as the preferred specification** and report Blundell-Bond alongside it as a robustness comparison. Both are estimated in two steps with Windmeijer-corrected standard errors, and the instrument set is collapsed to keep it from proliferating with the time dimension.

The two estimators are run on different numbers of observations, which is mechanical rather than a data discrepancy. The Arellano-Bond estimator uses only the *differenced* equation, and differencing consumes the first available period for each country (a difference needs both t and $t-1$). The Blundell-Bond estimator instead stacks **two** equations, the differenced equation *and* the levels equation, so its observation count combines both and recovers the first-period information that differencing discards, leaving it with roughly twice as many observations. The larger count reflects the system estimator's additional moment conditions, not a larger underlying sample; the exact figures are reported with the results in Section 7.

Instrument validity is assessed with the **Sargan/Hansen** test of over-identifying restrictions and the **AR(2)** test for second-order serial correlation in the differenced residuals; in both, a p-value above 0.10 indicates that the instruments are valid. The estimation uses complete cases (observations with the lagged dependent variable and all four controls present), about 1,500 country-periods before the GMM transformations described above.

7. Results

7.1. The Metric Matters: LCRPGR vs. BpCR

We first compare the two metrics in the static (lag-free) two-way fixed-effects model, holding the data, controls, and fixed effects identical so that any difference is due to the dependent variable alone. To keep every specification comparable, all models in this section, static and dynamic, are estimated on the common sample for which the lagged dependent variable and all four controls are available; the static models therefore report the same number of observations as the dynamic ones (1,499), so that differences across columns reflect the specification and not a change in the sample. The contrast between the metrics is stark (Table 4).

Table 4: Static two-way fixed-effects model on the estimation sample (1985-2020), same sample and controls for both metrics. Coefficient with significance stars; clustered standard errors in parentheses.

Variables	LCRPGR (log)	BpCR
ln(Urban Density)	-2.6079	-0.0218***
	(1.9467)	(0.0070)
Net Migration (% Pop)	-0.0239	-0.0008**
	(0.0392)	(0.0003)
ln(GDP per capita)	+2.7401	-0.0075***
	(1.6817)	(0.0028)
Urban Pop. Share	-14.9322	-0.0014
	(16.1412)	(0.0261)
Observations	1,499	1,499
Within R^2	0.0033	0.0647
Country FE	Yes	Yes
Period FE	Yes	Yes

*** p<0.01, ** p<0.05, * p<0.1.
Clustered (by country) standard errors in parentheses.

With the official LCRPGR as the dependent variable, the model is essentially uninformative: the within R^2 is only 0.003 and not a single control is statistically significant, because the metric's arithmetic tail outliers swamp any systematic variation. Replacing the dependent variable with BpCR, and changing nothing else, raises the within R^2 to 0.065, roughly 19 times higher, and the coefficients become significant and correctly signed: denser cities, richer countries, and higher in-migration are all associated with less land per capita. This is the clearest evidence for H1: the choice of metric, not the data, determines whether the relationships can be detected at all.

7.2. *The Final Dynamic Model*

Table 5: Final dynamic-panel model: FE stepwise build-up, Arellano-Bond (difference) and Blundell-Bond (system) GMM. Estimate with significance stars on top, clustered/robust standard error in parentheses below.

Variables	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t−1]	+0.2535***	+0.2346***	+0.2346***	+0.2286***	+0.2287***	+0.6057***	+0.5908***
	(0.0578)	(0.0582)	(0.0585)	(0.0583)	(0.0583)	(0.1081)	(0.0623)
ln(Urban Density)		-0.0147**	-0.0146**	-0.0169**	-0.0165**	-0.0523***	-0.0021**
		(0.0064)	(0.0065)	(0.0065)	(0.0065)	(0.0142)	(0.0010)
Net Migration (% Pop)			-0.0008**	-0.0008**	-0.0008**	-0.0005*	-0.0010***
			(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0002)
ln(GDP per capita)				-0.0068**	-0.0068**	-0.0138**	-0.0004
				(0.0028)	(0.0028)	(0.0060)	(0.0003)
Urban Pop. Share					-0.0031	-0.0310	+0.0079*
					(0.0225)	(0.0520)	(0.0045)
Observations	1,499	1,499	1,499	1,499	1,499	1,351	2,895
Within R^2	0.076	0.091	0.108	0.125	0.125	–	–
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sargan (p)	–	–	–	–	–	0.44	0.14
AR(2) (p)	–	–	–	–	–	0.71	0.47

*** p<0.01, ** p<0.05, * p<0.1.
Clustered/robust (Windmeijer) standard errors in parentheses.

In the preferred Arellano-Bond specification the lagged term is large and significant ($\rho \approx 0.61$): about three-fifths of one period's BpCR persists into the next. The GMM estimate is roughly three times the fixed-effects estimate ($\rho \approx 0.23$), exactly the direction the Nickell bias predicts, and confirms **strong path-dependence** in per-capita land use (H5). Among the drivers, **urban density** enters negative and highly significant, a compact-city effect (H2); **net migration** is negative, consistent with in-migrants being absorbed into the existing stock (H4); and **GDP per capita** is negative and significant, so that within countries, getting richer is associated with **densification rather than sprawl** (H3). The urban population share is insignificant once the rest is controlled, suggesting that urban *form*, not the stage of urbanisation, drives land use.

As anticipated in Section 6, the two GMM estimators report different observation counts: the Arellano-Bond (difference) estimator uses 1,351 observations, while the Blundell-Bond (system) estimator reports 2,895. The difference is purely mechanical: the system estimator stacks the differenced and levels equations and so recovers the first-period information that differencing discards, roughly doubling the count, rather than drawing on any additional data. Both estimators pass the diagnostic tests: the Sargan/Hansen test does not reject the over-identifying restrictions and the AR(2) test finds no second-order serial correlation (both $p > 0.10$), supporting the validity of the instruments. The Blundell-Bond estimates are close to Arellano-Bond in sign and significance, indicating the findings are not an artefact of the chosen estimator.

7.3. *Hypotheses*

Taken together, the results support all five hypotheses. The metric comparison supports **H1** (BpCR is the more reliable, better-fitting metric); the dynamic model supports **H2** (density reduces land per capita), **H3** (income reduces it, within countries), **H4** (in-migration densifies), and **H5** (per-capita land use is strongly path-dependent).

8. Robustness

The findings are stable across a 2000–2020 subsample, a restriction to shrinking-population countries, and alternative instrument depths; the Sargan and AR(2) diagnostics pass throughout, supporting instrument validity.

9. Discussion

The within-country result that income is associated with densification may appear to contradict the cross-sectional literature in which richer countries have historically sprawled. The two are complementary: *across* countries, higher income has coincided with more sprawl historically, but *within* a country, rising income over time is associated with denser, not more sprawling, urban land use. Measuring this distinction requires a metric that is stable within country-period – which BpCR provides and LCRPGR does not.

10. Conclusion

The official SDG 11.3.1 indicator is statistically unstable by construction. The Built-up per Capita Rate is a drop-in, log-based alternative that is finite, symmetric, and interpretable for every country-period, and it reproduces the policy signal without the instability. Estimated dynamically on a global panel, per-capita urban land use is strongly path-dependent and, within countries, declines with density, in-migration, and income. The contribution is as much **methodological** – a stable metric and a bias-corrected dynamic estimator – as substantive, and it complements rather than overturns the existing literature.

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